

SUMMER EXAMINATION 2026

Artificial Intelligence

BSIT - 3rd Semester Time: 90 Minutes Max Marks: 80

Name: _____ Roll No.: _____ Date: _____

1. AI is best described as the study of agents that perceive and act
(A) intelligent behavior (B) database normalization
(C) file compression (D) packet framing
2. The Turing test evaluates whether a machine can
(A) converse indistinguishably from a human (B) sort numbers
(C) store facts (D) encrypt files
3. Weak AI is designed for
(A) a limited task (B) every intellectual task
(C) only robotics (D) only games
4. A rational agent chooses an action that maximizes
(A) expected performance (B) memory usage
(C) program length (D) randomness
5. In PEAS, the E stands for
(A) Environment (B) Execution
(C) Encoding (D) Evaluation
6. A fully observable environment provides the agent with
(A) complete relevant state information (B) only rewards
(C) no sensors (D) past actions only
7. A reflex agent selects actions mainly from
(A) the current percept (B) a learned value function
(C) a planning graph (D) a proof tree
8. A model-based agent maintains
(A) an internal state (B) only a lookup table
(C) a random seed (D) a utility invoice
9. A utility-based agent uses utility to
(A) rank desirable outcomes (B) tokenize text
(C) prune syntax trees (D) count database rows
10. An episodic task environment has actions that
(A) do not affect later episodes (B) always require planning
(C) must be continuous (D) have no observations
11. Breadth-first search expands the
(A) shallowest frontier node (B) lowest heuristic node
(C) deepest node (D) random node
12. Depth-first search uses a frontier organized as a
(A) stack (B) queue
(C) priority table (D) hash ring
13. Uniform-cost search selects the frontier path with
(A) lowest accumulated cost (B) lowest depth
(C) highest heuristic (D) most children
14. A heuristic function estimates
(A) cost from a state to a goal (B) the exact training label
(C) network bandwidth (D) number of agents
15. A star search evaluates a node with
(A) $f(n)=g(n)+h(n)$ (B) $f(n)=g(n)-h(n)$
(C) $f(n)=h(n)/g(n)$ (D) $f(n)=g(n)h(n)$ only
16. A heuristic is admissible when it
(A) never overestimates true remaining cost (B) always equals true cost
(C) is negative (D) ignores the goal
17. If every edge costs 1, a path of depth 4 has cost
(A) 4 (B) 1
(C) 0 (D) 8
18. Hill climbing can fail at a
(A) local maximum (B) root node only
(C) terminal compiler (D) balanced database
19. Simulated annealing sometimes accepts
(A) a worse move (B) only a goal move
(C) no move (D) a duplicate state
20. Minimax assumes that the opponent
(A) acts optimally (B) acts randomly
(C) never moves (D) maximizes our score
21. Alpha-beta pruning changes minimax results
(A) never (B) only at even depth
(C) always (D) only with chance nodes
22. In alpha-beta pruning, alpha is the best value found for
(A) MAX so far (B) MIN so far
(C) the root heuristic (D) a random leaf
23. A constraint satisfaction problem consists of variables, domains, and
(A) constraints (B) rewards
(C) gradients (D) tokens
24. Backtracking search usually assigns
(A) one variable at a time (B) all variables randomly
(C) only goal variables (D) no values
25. The MRV heuristic chooses the variable with
(A) fewest legal values (B) largest domain
(C) highest cost (D) most neighbors only
26. Arc consistency removes values that
(A) have no supporting neighbor value (B) are globally optimal
(C) are duplicated (D) are outside the root
27. In STRIPS, an action has preconditions and
(A) effects (B) pixels
(C) gradients (D) priors
28. Forward planning starts at
(A) the initial state (B) the goal state
(C) the deepest leaf (D) the utility maximum
29. Backward planning works backward from
(A) the goal (B) the first percept
(C) the largest domain (D) the reward table
30. A proposition is a statement that is
(A) true or false (B) always numerical
(C) a probability distribution (D) a neural layer
31. Forward chaining applies rules from
(A) known facts toward conclusions (B) goals toward facts
(C) random facts (D) only contradictions
32. Backward chaining begins with
(A) a query goal (B) an empty database
(C) a sensor reading (D) a heuristic value
33. A Bayesian network represents dependencies using a
(A) directed acyclic graph (B) stack
(C) confusion matrix (D) parse tree only
34. Naive Bayes assumes features are conditionally independent given the
(A) class (B) learning rate
(C) action (D) cluster index
35. Fuzzy logic allows truth values
(A) between 0 and 1 (B) only -1 and 1
(C) only 0 or 1 (D) above 100
36. Supervised learning uses
(A) labeled examples (B) no observations
(C) only rewards (D) only rules
37. Unsupervised learning seeks structure in
(A) unlabeled data (B) test labels
(C) utility functions (D) action policies
38. Reinforcement learning learns from
(A) rewards and penalties (B) class labels only
(C) SQL joins (D) grammar rules
39. A validation set is used mainly for
(A) model selection and tuning (B) final unbiased reporting
(C) data collection (D) encryption
40. Overfitting means a model fits training data well but
(A) generalizes poorly (B) has no parameters
(C) cannot train (D) has low variance always

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1. High bias commonly causes
(A) underfitting (B) memorization
(C) data leakage (D) high variance only
2. Precision is the fraction of predicted positives that are
(A) actually positive (B) actually negative
(C) all observations (D) missed positives
3. Recall is the fraction of actual positives that are
(A) correctly found (B) predicted negative
(C) all negatives (D) true negatives only
4. For TP=8 and FP=2, precision is
(A) 0.80 (B) 0.20
(C) 0.60 (D) 0.90
5. The F1 score is the harmonic mean of
(A) precision and recall (B) accuracy and loss
(C) bias and variance (D) speed and memory
6. A perceptron computes a weighted sum followed by
(A) an activation (B) a database join
(C) a graph search (D) a token split
7. Gradient descent updates parameters using the
(A) negative loss gradient (B) positive labels only
(C) largest feature (D) random action
8. A convolutional neural network is especially suited to
(A) spatial patterns (B) symbolic proofs only
(C) transaction logs only (D) shortest paths
9. Pooling in a CNN commonly reduces
(A) spatial dimensions (B) the number of classes
(C) training labels (D) token vocabulary only
10. An RNN is designed to process
(A) sequences (B) static tables only
(C) graphs only (D) images without order
11. An LSTM uses gates to help preserve
(A) long-range dependencies (B) database keys
(C) pixel colors only (D) search depth
12. A Markov decision process includes states, actions, transition model, rewards, and
(A) a discount factor (B) a compiler
(C) a parse tree (D) a domain vocabulary
13. A policy maps states to
(A) actions or action probabilities (B) class labels only
(C) database rows (D) heuristic costs
14. Q-learning estimates
(A) action values (B) class priors only
(C) parse probabilities (D) pixel edges
15. The Q-learning update uses the reward plus discounted
(A) next-state best Q value (B) current input size
(C) training accuracy (D) heuristic depth
16. Exploration in reinforcement learning means
(A) trying uncertain actions (B) always choosing the best known action
(C) removing rewards (D) stopping training
17. A false positive occurs when a negative case is
(A) predicted positive (B) predicted negative
(C) left unlabeled (D) counted twice
18. Data leakage occurs when training uses information
(A) unavailable at prediction time (B) from a larger batch
(C) with missing values (D) after normalization
19. Responsible AI emphasizes fairness, transparency, privacy, and
(A) accountability (B) unlimited surveillance
(C) opaque decisions (D) forced automation
20. A model card is intended to document
(A) model purpose, limits, and performance
(B) network routes
(C) database indexes (D) source code licenses only
21. Differential privacy protects individuals by adding controlled
(A) noise (B) labels
(C) search depth (D) network hops
22. A vector dot product measures alignment and is largest for vectors pointing
(A) in similar directions (B) in opposite directions
(C) at random (D) with zero length only
23. The determinant of a 2 by 2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is
(A) $ad-bc$ (B) $ab+cd$
(C) $ac-bd$ (D) $a+b+c+d$
24. If a fair binary event has probability 0.5, its odds of success are
(A) 1 to 1 (B) 1 to 2
(C) 2 to 1 (D) 0 to 1
25. Cross-entropy penalizes a confident prediction that is
(A) wrong (B) correct
(C) uniform (D) missing a feature
26. A confusion matrix summarizes
(A) classification outcomes (B) regression slopes
(C) search paths (D) cluster centroids
27. A ROC curve plots true positive rate against
(A) false positive rate (B) true negative rate
(C) loss (D) sample size
28. K-means assigns points to the nearest
(A) centroid (B) decision stump
(C) support vector (D) parse node
29. PCA projects data onto directions of greatest
(A) variance (B) class count
(C) reward (D) text frequency only
30. The principal purpose of normalization in ML is to
(A) put features on comparable scales (B) create labels
(C) increase leakage (D) remove all outliers
31. A learning rate that is too large may cause training to
(A) diverge (B) stop overfitting automatically
(C) produce labels (D) become deterministic
32. A learning rate that is too small usually makes convergence
(A) slow (B) impossible in every case
(C) random (D) independent of data
33. An agent's sensors provide percepts, while its actuators produce
(A) actions (B) heuristics
(C) constraints (D) utilities
34. Iterative deepening combines the low memory of DFS with the completeness of
(A) BFS (B) hill climbing
(C) minimax (D) beam search
35. A goal test checks whether a state satisfies
(A) the goal condition (B) the heuristic estimate
(C) the action cost (D) the percept history
36. Knowledge engineering converts domain expertise into
(A) a knowledge base (B) a pixel matrix
(C) a routing table (D) a loss curve
37. The utility of an outcome represents its
(A) desirability (B) token count
(C) search depth (D) probability only
38. An expert system usually contains a knowledge base and an
(A) inference engine (B) image encoder
(C) order queue (D) optimizer
39. The discount factor gamma is normally constrained to the interval
(A) 0 to 1 (B) 1 to 2
(C) -2 to -1 (D) 10 to 100
40. An AI agent's performance measure specifies
(A) how success is evaluated (B) how files are compressed
(C) how tokens are split (D) how keys are stored